

THE IMPACT OF MONETARY POLICY ON HOUSING PRICES IN CHINA

Shen Chen

Western Michigan University

shen.chen@wmich.edu

Wan Wei

Arkansas Tech University

wwei@atu.edu

Peng Huang

Arkansas Tech University

Phuang3@atu.edu

ABSTRACT

This paper examines the impact of monetary policy on housing prices in China with a VAR model. Granger causality tests, impulse response functions, and variance decompositions are used to analyze the impacts of two monetary policy variables, market-based short-term interest rates and money supply, on housing prices. The results show that a contractionary monetary policy will cause the growth rate of housing prices to decline in China. In particular, a positive shock to market-based interest rates measured by the 7-day interbank offered rate has a significant and negative effect on housing prices in a range from five months to one and a half years after the shock takes place. However, our paper finds no evidence that supports the significant impact from money supply on housing prices. The results of our paper imply that the market-based short-term interest rates are effective monetary policy instruments for the central bank in China to conduct its policy to affect housing prices.

Keywords: *Housing Prices, Monetary policy, China*

JEL Classification: E52, G12

1. INTRODUCTION

Many empirical studies find that housing prices are affected by monetary policy. In particular, after the 2008-2009 U.S. subprime crisis, the Federal Reserve has been under attack for its loose monetary policy during the years preceding the crisis. Some research, such as Taylor (2007), Bernanke (2009), and Holt (2009), believe that an aggressive and expansionary monetary policy including abundant liquidity and low interest rates was probably the most important macroeconomic driver in the formation of the bubble in the U.S. housing market before 2008. Meanwhile, studies such as Iacoviello and Minetti (2003) and Iacoviello (2005) assert that a contractionary monetary policy could lead to a decrease in housing prices in European countries and the U.S.

The focus of our paper is to study whether a monetary policy may affect housing values with data from China. The housing market in China has experienced a rapid boom since the privatization and commoditization of residential real estate in 1998. For example, by the end of 2009, the total value of the residential real estate in China climbed to 91.5 trillion Yuan (13.4 trillion US dollars), which was about three times the nominal GDP of China in the same year according to China Securities Journal (2009). Although the primary objective of the central bank in China, the People's Bank of China (the PBC), is to promote economic growth and foster low and stable inflation, the fast increase of the real estate values have caused policymakers to become increasingly worried about a potential risk of the housing bubble in China.

In the last two decades, the PBC has implemented both expansionary and contractionary monetary policies. For example, in order to combat the economic slowdown during the 2008-2009 global financial crisis, the PBC adopted a highly expansionary monetary, which included a tremendous increase in money supply, bank loans, and a series of cuts in interest rates. The PBC also lowered the minimum down payment for home purchases. The housing market seemed to respond strongly and favorably to this expansionary monetary policy. According to Xu and Chen (2012), the national home price growth index rebounded in a short period of time, from -1.1% in the first quarter of 2009 to 5.8% in the fourth quarter of 2009. Starting from the second half year of 2009, the PBC took a series of actions, which included raising the reserve ratio and interest rates to

tighten its monetary policy and increasing the minimum down for home purchases. Again, it appears that the housing market responded to this contractionary monetary policy. For instance, the volume of real estate transactions declined significantly in 2010 (Xu and Chen, 2012). Meanwhile, the monthly growth rate of housing prices slowed down from 2.00% at the beginning of 2010 to nearly zero percent in the middle of 2010. With a slight rebound at the end of 2010, the housing prices started to decline in the middle of 2011 until the middle of 2012.

Our paper attempts to examine the influence of monetary policy on housing prices in China with a vector autoregression model (VAR). We focus on investigating two monetary policy variables: short-term interest rates and money supply. Granger causality test, impulse response functions, and variance decompositions are employed to analyze these impacts. Our results show a contractionary monetary policy does exert a downward pressure on housing prices in China. In particular, a positive interest rate shock, that is an increase in short-term interest rates, has a significant and negative impact on housing prices. However, a shock to money supply does not have an influence on the housing prices.

The remainder of this paper is organized as follows: Section 2 is a literature review, Section 3 introduces the model used in the paper, Section 4 describes the data, Session 5 presents the empirical results, and the final section concludes.

2. LITERATURE REVIEW

Bernanke and Gertler (1995) argued that there are two channels through which monetary policy affects housing markets. The first channel is a household income channel. For example, a contractionary monetary policy shock, such as an increase in interest rates, can cause household income to decrease. A decrease in household income reduces household investment in a housing market, which in turn reduces housing prices. The second channel is a bank-lending channel. When interest rates increase, the cost of buying a house increases as well. As a result, the demand for housing from households goes down, which causes housing prices to drop.

A number of empirical studies have been conducted to study how a monetary policy may affect housing values. Many of their results are in favor of theoretical prediction from Bernanke and Gertler (1995). Tsatsaronis and Zhu (2004) found that housing prices increase strongly in response to an interest rate cut with data from 17 industrialized countries. Using three European countries, Iacoviello and Minetti (2003) found that housing price responds negatively to a change in interest rate. Likewise, Iacoviello (2005) used data from the U.S. and found a negative impact of interest rate on housing pricing as well.

Additionally, several studies have been conducted to examine how a monetary policy affects the housing market in China. Koivu (2012) demonstrated that an expansionary monetary policy results in higher housing prices in China with data from 1998-2008. James and Hui (2010) used data from 1999 to 2010 to show that a tightening monetary policy reduces property prices. Zhang, Hua and Liang (2011) used data from 1999 to 2010 and found that the monetary policy variables are key factors in influencing housing prices in China. Xu and Chen (2012) showed that an expansionary monetary policy tends to accelerate the housing price growth while a contractionary monetary policy tends to decelerate the housing growth. Although most of existing studies show a significant impact of monetary policy on housing prices in China, Yao, Luo and Loh (2013) claimed that the monetary policy conducted by the PBC has little impact on housing prices by using data from 2005 to 2010. They attributed their finding to investors' irrational and speculative behavior. Similar to Yao, Luo and Loh (2013), we also examine the direction, magnitude, and timing of the effects of monetary policy on housing prices in our study.

We are particularly interested in investigating whether or not two monetary policy variables, short-term interest rates and money supply, affect housing prices in China. We choose the 7-day interbank offered rate as a representation of market-based short-term interest rates. There are several reasons to use the 7-day interbank offered rate as a monetary policy instrument in China. First, the PBC publicly states that an interest rate policy is an important part of its monetary policy. Second, unlike the Federal Reserve, the PBC does not target a specific interest rate as its primary monetary instrument. However, the PBC conducts its open market operations by issuing PBC bills. Laurens and Maino (2007) asserted that there exists a stable relationship between the PBC bills rates and short-term interbank offered rates. They claimed that the PBC controls short-term interest

rates to control liquidity conditions in the interbank market. Therefore, the market-based interest rates have become more and more important as interest rate liberalization continues in China. Third, among the market interest rates, the 7-day interbank offered rate has been long and actively used as a benchmark for other interest rates and financial products. More specifically, it serves as the China Interbank Offered Rate (CHIBOR) market's benchmark rate.

The money supply used in our study is measured by M2, which is a broad monetary aggregate. We follow Koivu (2012) and Yao, Luo and Loh (2013) and use M2 as a monetary policy variable in our model. The PBC has publicly announced money supply indicators including M0, M1, and M2 since 1994. More importantly, the PBC established the money supply as the single intermediate target for its monetary policy in 1998 (Laurens and Maino, 2007). Koivu (2012) also argued that the PBC typically set growth rates for M2 and closely monitors the movements of M2.

3. EMPIRICAL MODEL

To study the impacts of monetary policy on housing price, we employ a structural VAR model with six system variables: housing price index (House), industrial added value (Y), consumer price index (CPI), money supply growth rate (M2), short-term interest rate (R), and real exchange rate (EX). The VAR model allows all the six system variables to influence each other endogenously. Among these system variables, money supply (M2) and short-term interest rate (R) are treated as monetary policy variables. A basic reduced-form VAR process is given as: $y_t = A(L)y_t + u_t$, where $y_t = (y_{1t}, y_{2t}, \dots, y_{nt})$ contains N endogenous variables and L is the lag term, $A(L) = A_1L + A_2L^2 + \dots + A_iL^i$, and u_t is an N-dimensional process and assumed to be serially uncorrelated.

In this paper, we want to model the contemporaneous relations between the variables. Hence, we need a structural form of the VAR model:

$$Ay_t = A^*(L)y_t + B\varepsilon_t = A_1^*y_{t-1} + A_2^*y_{t-2} + \dots + A_i^*y_{t-i} + B\varepsilon_t$$

where the structural errors ε_t are assumed to be serially and cross-sectionally uncorrelated. We can obtain the relationship between the error terms of system variables in the reduced and structural

forms: $u_t = A^{-1}B\varepsilon_t$. Then, we can have $Au_t = B\varepsilon_t$ and decompose the structural errors ε_t . In our case, the contemporary relations are as following:

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ * & 1 & 0 & 0 & 0 & 0 \\ * & * & 1 & 0 & 0 & 0 \\ * & * & * & 1 & 0 & 0 \\ * & * & * & * & 1 & 0 \\ * & * & * & * & * & 1 \end{bmatrix} \begin{bmatrix} u_t^Y \\ u_t^{CPI} \\ u_t^{M2} \\ u_t^R \\ u_t^{EX} \\ u_t^{HP} \end{bmatrix} = \begin{bmatrix} b_{11} & 0 & 0 & 0 & 0 & 0 \\ 0 & b_{22} & 0 & 0 & 0 & 0 \\ 0 & 0 & b_{33} & 0 & 0 & 0 \\ 0 & 0 & 0 & b_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & b_{55} & 0 \\ 0 & 0 & 0 & 0 & 0 & b_{66} \end{bmatrix} \begin{bmatrix} \varepsilon_t^Y \\ \varepsilon_t^{CPI} \\ \varepsilon_t^{M2} \\ \varepsilon_t^R \\ \varepsilon_t^{EX} \\ \varepsilon_t^{HP} \end{bmatrix}$$

where * indicates the parameter that is estimated in the system. Following Vargas-Silva (2008), we assume that the industrial added value adjusts slowly to the shocks of the other system variables and it is only allowed to react to the shocks with a lag. The inflation variable is allowed to react immediately to shocks to output proxied by the industrial added value. Furthermore, non-monetary policy variables (e.g. the output and inflation) and monetary policy variables (e.g. money supply and short-term interest rates) can affect housing prices contemporarily. Thus, the housing price is allowed to respond immediately to shocks to all the other system variables.

4. DATA

The data of the housing price index and the industrial added value growth rate are collected from Statistical Yearbook of China. The inflation rate is calculated using the consumer price index, which is also found in the Statistical Yearbook of China. The interest rate is measured by the 7-day interbank offered rate and the money supply is measured by M2, both of which come from the People's Bank of China. The data of the real exchange rate can be found in the World Bank database. All variable data is monthly, dating from July 2005 to February 2014.

Table 1 shows the statistical description of the variables used in our paper. According to Table 1, the growth rate of the housing price index is in a range between -0.70% and 1.90%. In addition, the two monetary policy variables we focus on, the 7-day interbank offered rate and the growth rate of M2, fluctuate in a wide range. The 7-day interbank offered rate fluctuates between 1.00% and 7.08% and the growth rate of M2 fluctuates between -0.60% and 3.58% in our sample period.

The housing price index has been released by National Bureau of Statistics of China since 2005. It is a commonly used housing price index in China and it covers 70 major cities in China. A growth rate of the housing price index is computed as the monthly growth rate of the housing price index. This index has also been used by Ahuja et al. (2010). The industrial added value growth rate is measured by the growth rate of gross industrial output value in China. We used this variable to represent the real output growth rate in China. The real exchange rate is measured by real effective exchange rates of the Chinese Yuan. It is measured by the Bank for International Settlements (BIS) effective exchange rates with 2010 as the base year.

Figure 1 displays the movements of the system variables used in our paper. We can see that the 7-day interbank offered rate fell significantly and M2 increased sharply in the second half of 2008. This demonstrated that an aggressive expansionary monetary policy was conducted by the PBC to fight against the impact of the global financial crisis discussed previously. The housing market appeared to respond to this expansionary monetary policy. The growth rate of the housing price index began rebounding strongly after the growth rate hit the bottom at the end of 2009. It is worth noting that starting from the middle of 2009, the 7-day interbank offered rate started to rise from the lowest level around 1% and the growth rate of M2 declined from the peak of around 3.6%, which signaled the beginning of a tightening monetary policy by the PBC to curb the fast-growing asset prices. By the end of our sample period, the 7-day interbank offered rate increased to about 5% and the growth rate of money supply reduced to about 1%. Again, this contractionary monetary policy seemed to have an impact on the real estate market. The growth rate of the housing price index decreased sharply from nearly 2% to 0% in the first half of 2010. Overall, Figure 1 gives us some hints that the monetary policy does have an influence on the housing prices in China.

5. EMPIRICAL RESULTS

5.1 Unit Root and Cointegration

We used an Augmented Dickey-Fuller test to examine whether or not the system variables in our model have a unit root. The test results indicate that M2, CPI, and EX have a unit root. Therefore,

these three variables are nonstationary. If more than one variable in the model has unit roots, they might be cointegrated. A cointegration implies a long-run equilibrium relationship that exists among non-stationary variables. We then used a Johansen cointegration test to check whether there is a cointegration among M2, CPI, and EX. The outcome of the Johansen test showed that no cointegration exists among these variables at a 10% significant level.

5.2 Granger Causality Test

In order to check whether there is a causal relationship between the housing price and other system variables, we utilized a Granger causality test. The lag length we selected for the variables is two, which is based on the Akaike Information Criterion (AIC). Furthermore, we conducted a VAR residual serial correlation Lagrange multiplier (LM) test. The results indicate that there is no autocorrelation up to lag 11, which means our model fits the data well.

Table 2 reports the Granger causality test results. With the exception of the interest rate, the p values of all the other variables are greater than 10%. This means the industrial added value growth rate, inflation rate, money supply, and exchange rate do not Granger-cause the housing prices. Note that the money supply is a monetary policy variable in our model. The result of our causality test suggests that the growth rate of the money supply has no impact on housing prices. By contrast, the p-value of the interest rate is less than 5%, which indicates that the interest rate does Granger-cause the housing price. Simply put, a change in the interest rate will result in a change in the housing price. The interest rate used in our model is the 7-day interbank offered rate that serves as a proxy for a monetary policy instrument used by the PBC. The results from our Granger causality test implies that a change in the 7-day interbank offered rate as a monetary policy instrument will result in a change in the housing prices in China.

5.3 Impulse Response Function

We also constructed a VAR model based off a Cholesky decomposition to analyze the impacts of the monetary policy shocks on housing prices. The impulse response functions are able to describe the response of a variable such as the housing price to a one standard deviation innovation of another variable for a period of 48 months. Based on the AIC and Schwarz Information Criterion (SIC), the lag length of our VAR model is 2.

Figure 2 illustrates the responses of the housing price to shocks of all the other system variables based on the impulse response functions. The solid line depicts the movement of the housing price and the two dash lines represent the confidence interval with two standard deviations. From Figure 2, we can see that a positive shock to the interest rate causes the housing price to decline, but the initial decline is not statistically significant. The decline of the housing price becomes statistically significant around the fifth month and the significant effect persists until around eight months after the initial interest rate shock. The graph also shows that given one standard deviation shock to the interest rate, the growth rate of housing will fall by about 0.1%. However, the housing price does not respond to a shock to the rest of the system variables, except a shock to the growth rate of the industrial added value. In particular, the response of the housing price index to a shock of the money supply is not statistically significant throughout the entire 48 months. The result of the impulse response function demonstrates that the market-based 7-day interbank offered rate as a monetary policy instrument can significantly affect the housing prices. In contrast, the money supply as a monetary policy intermediate goal has no significant impact on the housing prices. Note that the result of the impulse response function is consistent with the result of the Granger causality test. The negative impact of the short-term interest rates on the housing prices in China supports the theory from Bernanke and Gertler (1995), who argued that a contractionary monetary policy will reduce the growth rate of housing prices.

5.4 Variance Decomposition

The variance decomposition is used to identify the proportion of the forecast error variance in the housing price that can be explained by shocks to all other system variables. Table 3 shows the outcome of the variance decomposition from 3 months up to 24 months. With the exception of the housing price itself, none of the five system variables can explain more than 10% of the forecast error variance of the housing price for the first 6 months. The portions of the variance of the housing price that can be explained by all these five non-housing variables are not statistically significant in the first 9 months. However, it is worth noting that the shocks to the interest rate tend to explain more portions of the variances of the housing price from the very beginning until 24 months, which increases from 0.62% to 18.34%. Particularly, starting from the ninth month, the interest rate accounts for the largest portion of the variance of the housing price among all the five

system variables, with the exception of the housing price itself. Moreover, from 12 months to 18 months, the portions of the forecast error variance of the housing price than can be explained by the interest rate are statistically significant at the 10% level. By contrast, the money supply as a monetary policy variable can barely capture 2% of the variance of the housing price throughout the entire period and none is statically significant. Again, the result of the variance decomposition indicates that in China the short-term interest rate as a monetary policy instrument is a key driver behind the changes of the housing prices, while the money supply has little explanatory power about the housing prices.

6. CONCLUSION

This study examines the impact of monetary policy on housing prices in China in terms of two monetary policy variables: the short-term interest rate and money supply. The Granger causality test, impulse response functions, and variance decomposition are used to analyze the impact based on a VAR model. The findings and contributions of this study have both academic and policy implications, which are summarized as follows.

First, our results show that the short-term interest rate proxied by the 7-day interbank offered rate is a key monetary policy variable that consistently exerts a significant impact on the housing prices in China. This implies that the 7-day interbank offered rate is an effective and reliable monetary policy instrument used by the PBC to deal with housing prices. Therefore, the PBC should rely more heavily on market-based instruments such as the 7-day interbank offered rate to conduct its monetary policy. Interest rate liberalization in China will help to enhance the effectiveness of the use of market-based monetary policy instruments by the PBC.

Second, the money supply measured by a monetary aggregate, M2, has no significant impact on housing prices in China. As mentioned above, money supply serves an intermediate goal for monetary policy conducted by the PBC. However, there are constant deviations between announced targets and actual values for money supply in China, which is due to an unstable relationship between money supply and real activity overtime (Laurens and Maino, 2007). Our

study further confirms that money supply may not serve as an effective monetary tool by the PBC to influence housing prices in China.

Third, our results exhibit the timing of the impact of interest rate on the housing market in China. The impulse response functions illustrate that a positive shock to the 7-day interbank offered rate significantly reduces the housing prices during a period of five months to nine months after the initial shock occurs. The variance decomposition function shows that the 7-day interbank offered rate is able to explain a statistically significant portion of the variance of the housing price over a window from one year to one and a half years. Both of the results imply that a contractionary monetary policy implemented by the PBC has no immediate impact on the housing prices in China. It will take about half a year for a contractionary monetary policy to start to influence the housing prices and the effect will last for approximately two years after the policy is initially implemented.

Lastly, our results are in line with theoretical predictions regarding the impacts of monetary policy on housing prices, that is, a contractionary monetary policy represented by a raise in interest rates tends to decrease the growth rate of the housing prices in China. In the future, we will continue to investigate how monetary policy affects housing prices in China by examining different monetary policy instruments such as the PBC bills rates. In addition to housing prices, we are also interested in investigating how various monetary policy instruments may affect other asset prices such as stock prices in China.

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TABLE 1
STATISTICAL DESCRIPTION OF DATA (07/2005 - 02/2014)

	Mean	S.D.	Min	Max
Y (%)	1.024	0.780	-1.38	4.60
CPI (%)	0.258	0.336	-0.65	1.29
R (%)	2.891	1.193	1.00	7.08
M2 (%)	1.351	0.656	-0.60	3.58
EX	99.463	10.093	83.79	120.59
House (%)	0.487	0.527	-0.70	1.90

Note: The industrial added value growth rate (Y) is measured by the growth rate of gross industrial output value in percentage. The inflation rate is measured by a percentage of the consumer price index (CPI). The growth rate of money supply is measured by the growth rate of M2 in percentage. The interest rate (R) is measured by the 7-day interbank offered rate. The real exchange rate is measured by the Bank for International Settlements (BIS) effective exchange rates. The housing price growth rate index (House) is measured by the growth rate of monthly housing prices in percentage.

TABLE 2
P-VALUES OF GRANGER CAUSALITY TEST

	Y	CPI	R	M2	EX
House	0.6298	0.3168	0.0138*	0.3678	0.2462

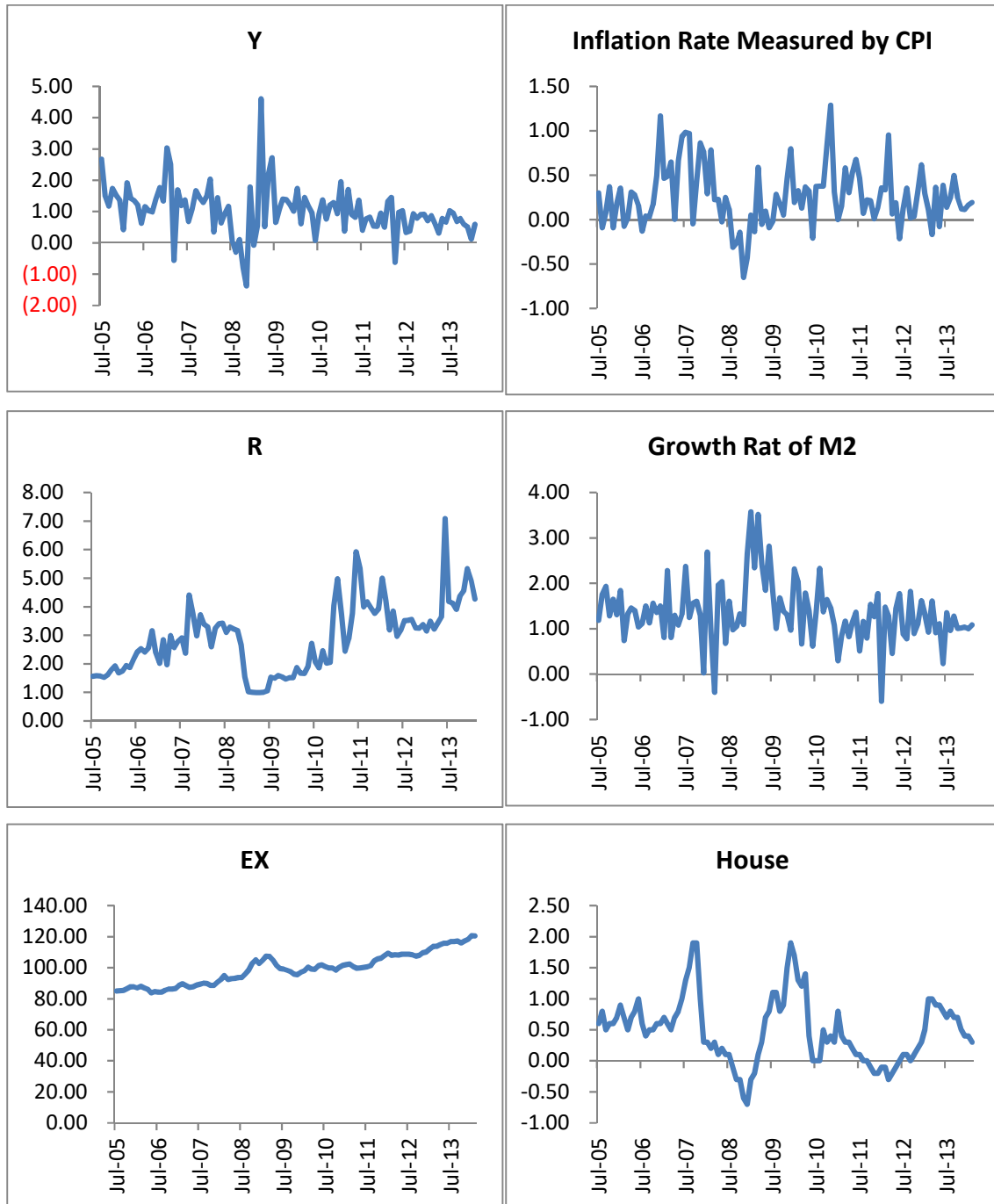
Note: This table reports the Granger causality test results. The numbers are the P-values. * indicates statistically significant at the 5% level; ** indicates statistically significant at the 10% level.

TABLE 3
VARIANCE DECOMPOSITION OF GROWTH RATE OF HOUSING PRICE INDEX

Period	Y	CPI	R	M2	EX	House
3	1.17	4.02	0.62	1.25	5.89	87.04
	(3.47)	(4.58)	(2.22)	(2.69)	(6.04)	(7.63)
6	3.41	8.68	5.70	2.01	7.06	73.14
	(5.58)	(8.60)	(5.68)	(4.23)	(6.87)	(11.40)
9	4.35	8.10	12.38	1.99	6.48	66.71
	(6.01)	(8.75)	(8.25)	(4.47)	(6.48)	(12.29)
12	4.41	8.44	16.28	1.88	6.15	62.84
	(5.93)	(8.22)	(9.46)**	(4.36)	(6.26)	(12.72)
15	4.31	9.38	17.78	1.82	6.03	60.68
	(5.92)	(8.13)	(10.14)**	(4.28)	(6.19)	(13.05)
18	4.24	10.04	18.22	1.79	6.01	59.70
	(5.98)	(8.29)	(10.67)**	(4.26)	(6.15)	(13.38)
21	4.21	10.36	18.32	1.79	6.00	59.32
	(6.03)	(8.45)	(11.13)	(4.25)	(6.11)	(13.70)
24	4.20	10.48	18.34	1.78	6.00	59.20
	(6.06)	(8.60)	(11.54)	(4.25)	(6.09)	(13.95)

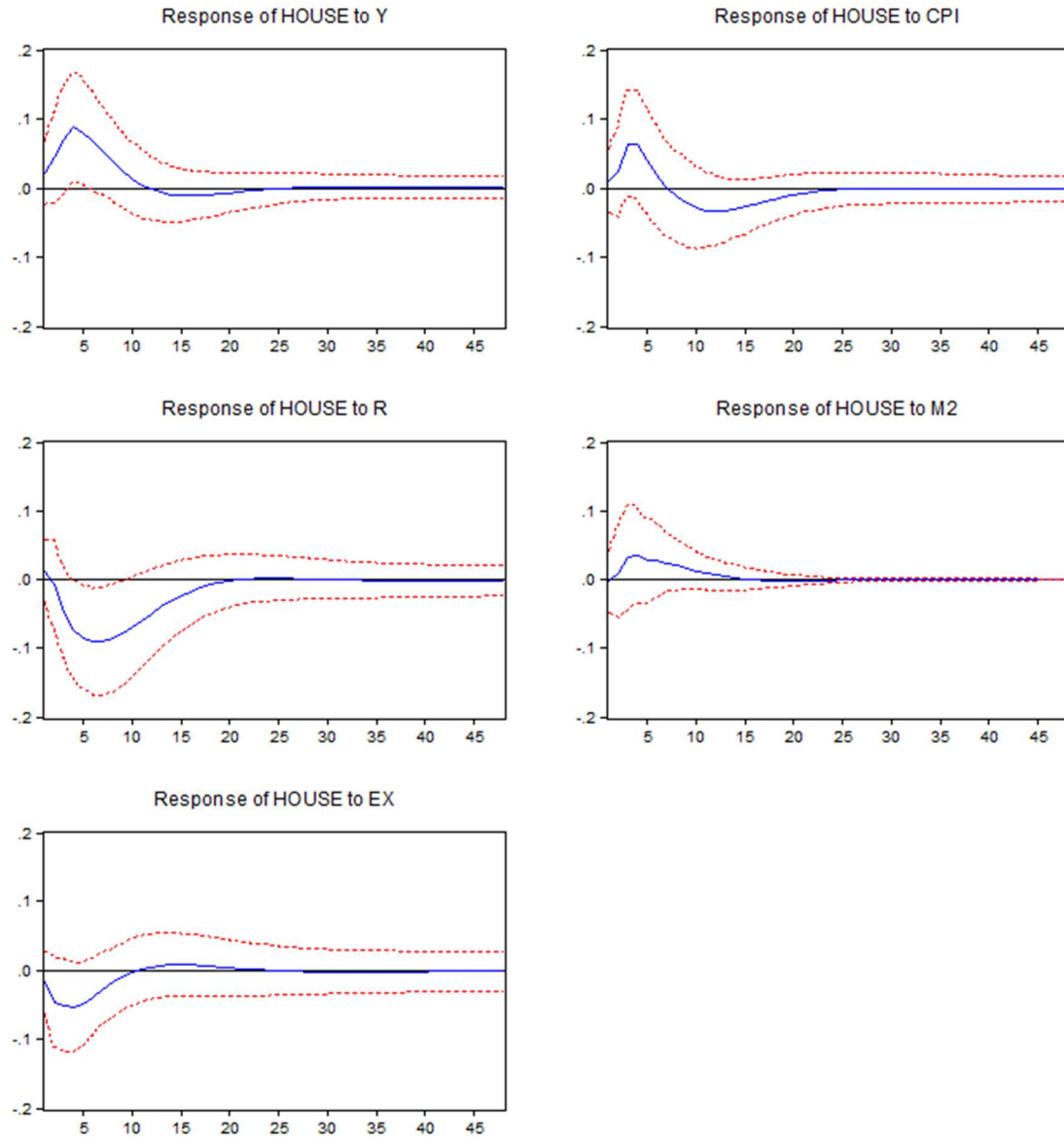
Note: This table reports the forecast error variance decomposition of the growth rate of housing price index on the first 24 months. Numbers in brackets are standard errors. * indicates statistically significant at the 5% level; ** indicates statistically significant at the 10% level.

FIGURE 1
PLOTS OF THE SYSTEM VARIABLES (07/2005 - 02/2014)



Note: All variable data is monthly, dating from July 2005 to January 2014. All the variables are expressed as percentage numbers except the exchange rate (EX).

FIGURE 2
IMPULSE RESPONSE ANALYSIS OF HOUSING PRICE INDEX



Note: Impulse response function of the growth rate of housing price index given one standard deviation positive shock to all the other system variables.